8. A technician changes an item's price

The technician updates the price of one location using maintenance mode.

Objective	• Allow a technician to change an item's price without changing the prices of other locations.	
Results	• The controller stores the new price and uses it when that location is selected during normal operation.	
Actors	• Technician, who understands the maintenance controls; vending machine controller.	
Related use cases	• Use Case 7: Initial setup. Use Case 1: Buying an item. Use Cases 5 and 6: Comparing payment with the item's price.	
Trigger	• The technician needs to change a price because the machine's quantity or pricing have changed.	
Stakeholders	• The owner and customers, who need the displayed and charged prices to be correct.	
Resources	• Programmed SoC Series V board, and Quartus.	
Assumptions	• A working setup with the seven-segment display, working money switches, working RAM set up for Items and the working keypad as well.	
Post-condition	• Location has a new price, and the controller has returned to normal operation. The updated price remains stored while power is maintained, unless another price change or reset occurs.	
Frequency	• Whenever the technician needs to update an item's price.	
Mission Priority	• High. Changing prices is a required function of the project.	
Safety Priority	• Should hold no safety concern	
Flow of events	**Step #**	**Action**
	1.	The technician confirms that no customer transaction is active.
	2.	The technician enters maintenance mode using the designated control. The controller indicates maintenance mode ad blocks customer purchases.

	3.	The technician selects location 11. The controller displays its current price of $1.35.
	4.	The technician enters a new price of $1.50, and the controller displays the entered value for review.
	5.	The technician uses the designated save control. The controller verifies that the location and price are valid, then updates location 11.
	6.	The technician exits maintenance mode.
	7.	The technician selects location 11 during normal operation and verifies that $1.50 is displayed. The technician also checks that the other locations' prices remain unchanged.
Alternate Events	**Step #**	**Action**
	3A	If the technician enters an invalid location, the controller rejects the selection and does not change any stored price.
	5A	If the entered price is below $0.05 or above $9.99, the controller rejects the entry and preserves the existing price.
	5B	If the technician cancels before saving, the controller discards the entered value and keeps the previous price.
	7A	If KEY0 is used to perform a power-up reset, the controller reloads the default prices. The maintenance change does not need to survive that reset.
Hazard/ Result /Mitigation	• If case is used incorrectly, the entire system is likely incorrect or not working to specifications.	
Requirements from Use Case	• Provide a maintenance mode for changing item prices. • Accept new prices from $0.05 through $9.99. • Update only the selected valid location after a save action. • Preserve existing prices when an entry is invalid or canceled before saving. • Display and use the new price during normal operation. • Retain changed prices while power is maintained, until another change or a power-up reset reloads the defaults.	